

Publications

Raphaël Tinarrage

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Journal Publications

- [1] Anton François and Raphaël Tinarrage. “Train-Free Segmentation in MRI with Cubical Persistent Homology”. In: *Journal of Mathematical Imaging and Vision* 68, 20 (2026). DOI: [10.1007/s10851-026-01300-1](https://doi.org/10.1007/s10851-026-01300-1).
- [2] Henrique Ennes and Raphaël Tinarrage. “LieDetect: Detection of Representation Orbits of Compact Lie Groups From Point Clouds”. In: *Foundations of Computational Mathematics* (2025), pp. 1–136. DOI: [10.1007/s10208-025-09728-4](https://doi.org/10.1007/s10208-025-09728-4).
- [3] Raphaël Tinarrage, Henrique Ennes, Lucas Resck, Lucas T Gomes, Jean R Ponciano, and Jorge Poco. “Empirical analysis of binding precedent efficiency in Brazilian Supreme Court via case classification”. In: *Artificial Intelligence and Law* (2025), pp. 1–67. DOI: [10.1007/s10506-025-09458-6](https://doi.org/10.1007/s10506-025-09458-6).
- [4] Raphaël Tinarrage, Jean R. Ponciano, Claudio D. G. Linhares, Agma J. M. Traina, and Jorge Poco. “ZigzagNetVis: Suggesting Temporal Resolutions for Graph Visualization Using Zigzag Persistence”. In: *IEEE Transactions on Visualization and Computer Graphics* 31.10 (2025), pp. 6852–6869. DOI: [10.1109/TVCG.2025.3528197](https://doi.org/10.1109/TVCG.2025.3528197).
- [5] Raphaël Tinarrage. “Recovering the homology of immersed manifolds”. In: *Discrete & Computational Geometry* 69.3 (2023), pp. 659–744. DOI: [10.1007/s00454-022-00409-5](https://doi.org/10.1007/s00454-022-00409-5).
- [6] Raphaël Tinarrage. “Computing persistent Stiefel–Whitney classes of line bundles”. In: *Journal of Applied and Computational Topology* 6.1 (2022), pp. 65–125. DOI: [10.1007/s41468-021-00080-4](https://doi.org/10.1007/s41468-021-00080-4).

Publications in Peer-Reviewed Conference Proceedings

- [7] Raphaël Tinarrage. “Simplicial Approximation to CW Complexes with Spherical Delaunay Triangulations”. In: *42nd International Symposium on Computational Geometry (SoCG 2026)*. Ed. by Hee-Kap Ahn, Michael Hoffmann, and Amir Nayyeri. Vol. 367. Leibniz International Proceedings in Informatics (LIPIcs). Dagstuhl, Germany: Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2026, 93:1–93:22. ISBN: 978-3-95977-418-5. DOI: [10.4230/LIPIcs.SoCG.2026.93](https://doi.org/10.4230/LIPIcs.SoCG.2026.93).
- [8] Beatriz Sabdin Chagas, Carla Marcondes Damian, and Raphaël Tinarrage. “O impacto da Súmula Vinculante 26 na diminuição de demanda similar no STF: uma análise quantitativa por modelos de machine learning”. Portuguese. In: *Direito Penal, Processo Penal e Criminologia: XI Encontro Internacional do CONPEDI Chile – Santiago*. Florianópolis, Brasil: Conselho Nacional de Pesquisa e Pós-Graduação em Direito (CONPEDI), 2022, pp. 82–103. ISBN: 978-65-5648-559-1. URL: <https://site.conpedi.org.br/publicacoes/129by0v5/gg2as8t1/Od71WWx2sWUgr61q.pdf>.

- [9] Ana Clara Macedo Jaccoud, Pedro Burlini de Oliveira, and Raphaël Tinarrage. “Progressão de regime em crimes hediondos no Supremo Tribunal Federal: uma análise empírica pela Súmula Vinculante 26”. Portuguese. In: *Direito Penal, Processo Penal e Criminologia: XI Encontro Internacional do CONPEDI Chile – Santiago*. Florianópolis, Brasil: Conselho Nacional de Pesquisa e Pós-Graduação em Direito (CONPEDI), 2022, pp. 399–418. ISBN: 978-65-5648-569-0. URL: <https://site.conpedi.org.br/publicacoes/129by0v5/502849so/6o53sVpwaxV5352U.pdf>.
- [10] Hirokazu Anai, Frédéric Chazal, Marc Glisse, Yuichi Ike, Hiroya Inakoshi, Raphaël Tinarrage, and Yuhei Umeda. “DTM-Based Filtrations”. In: *Topological Data Analysis: the Abel Symposium 2018*. Extended version of [11]. Springer. 2020, pp. 33–66. DOI: [10.1007/978-3-030-43408-3_2](https://doi.org/10.1007/978-3-030-43408-3_2).
- [11] Hirokazu Anai, Frédéric Chazal, Marc Glisse, Yuichi Ike, Hiroya Inakoshi, Raphaël Tinarrage, and Yuhei Umeda. “DTM-Based Filtrations”. In: *35th International Symposium on Computational Geometry (SoCG 2019)*. Ed. by Gill Barequet and Yusu Wang. Vol. 129. Leibniz International Proceedings in Informatics (LIPIcs). Dagstuhl, Germany: Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2019, 58:1–58:15. ISBN: 978-3-95977-104-7. DOI: [10.4230/LIPIcs.SoCG.2019.58](https://doi.org/10.4230/LIPIcs.SoCG.2019.58).

Preprints

- [12] Raphaël Tinarrage. *Sharp approximate Carathéodory theorem and application to iterated Delaunay refinement*. Extended version of Chapter 2 of [7]; invited for submission to the Special Issue of Selected Papers from SoCG 2026 of the [Journal of Computational Geometry](#). 2026. arXiv: [2606.25854](https://arxiv.org/abs/2606.25854) [[cs.CG](#)].